

The Stern–Gerlach experiment at 100

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100 years ago, Otto Stern and Walther Gerlach demonstrated that silver atoms have a quantized magnetic moment, as predicted from the Bohr–Sommerfeld model of the atom. But the correct interpretation of the result proved to be far more subtle — and revolutionary.

In a paper published in April 1922, Otto Stern and Walther Gerlach described an experiment in which they fired a narrow beam of silver atoms through a vertical magnetic field gradient. The atoms deposited on the target screen in two discrete beams, thus providing the first incontrovertible evidence that individual atoms can have a magnetic moment, and that the magnetic moment is quantized.

The publication had been hotly anticipated by many of the leading theorists of the day, from older ones like Albert Einstein and Paul Ehrenfest to the younger Wolfgang Pauli — some seeing it as the experiment that could make or break the recently developed quantum theory. Quantization of the magnetic moment had been predicted in extensions of Niels Bohr’s 1913 ‘semiclassical’ theory of atomic structure, although Stern’s preference clearly leaned towards disproving it. Quantum ideas did not agree with him, and together with his friend Max Laue, he had pledged that “if this nonsense of Bohr should in the end prove to be right, we will quit physics”¹.

In the end, Stern stayed in physics, despite his and Gerlach’s clear-cut result in favour of Bohr. But what the two physicists did not know at the time was that the magnetism of their silver atoms arose from an even more profound phenomenon, for which there was not yet a name or a theory: the intrinsic spin of the electron. As Wolfgang Pauli and Paul Dirac would explain a few years later, spin was an internal degree of freedom, the first of many to be discovered.

On its 100th anniversary, the Stern–Gerlach experiment maintains its iconic status and its theoretical intrigue. Better than any of the other foundational experiments of quantum physics, it incorporates virtually all of the profound features of the new theory — including superposition, entanglement, the uncertainty principle and the mysterious nature of measurement. It also continues to inspire philosophical discussions, technological advancements and modern re-enactments.

An elegant prediction

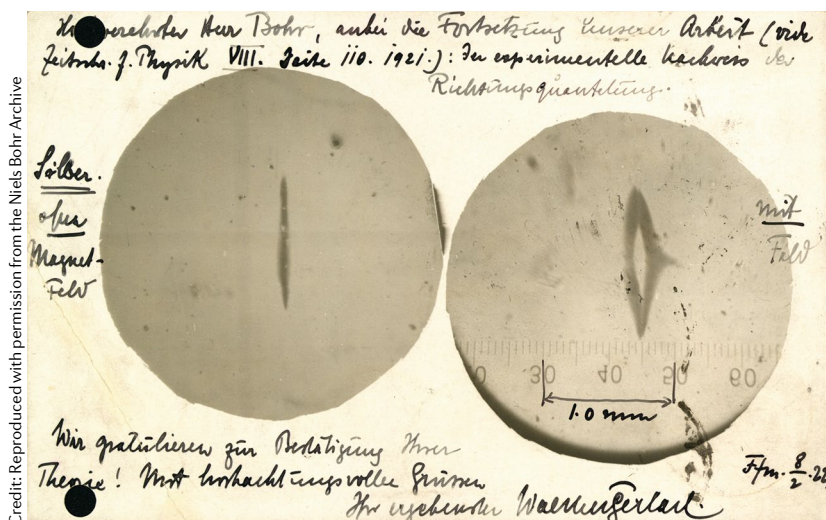
When Stern and Gerlach published their results², quantum mechanics as we know it had not yet been born. For two decades, several lines of evidence, including black-body thermodynamics, the photoelectric

effect and the spectral lines of the chemical elements had convinced leading theorists that classical physics was inadequate to describe matter and light at their smallest scales. But this was still the age of what later became known as the ‘old quantum theory’. The leading model was Bohr’s, which described electrons as having well-defined, classical trajectories that went around the nucleus in circles, but with energies confined to particular, quantized values. Bohr’s model, together with improvements by Arnold Sommerfeld and others, had proven itself by predicting with high accuracy the spectral series of atomic hydrogen — although attempts to extend that success to helium were a failure.

Sommerfeld’s development allowed for orbits to be elliptical and revealed that not only the electrons’ energies ought to take discrete values, but their angular momentum vectors should, too. (Sommerfeld had a peculiar interest in angular momentum, having co-authored a treatise on the spinning top — in four volumes.) This discretization had another, shocking, consequence. It seemed to mean that the orbital planes should only be found to be at particular angles. In English, the effect came to be called space quantization, but the original German term (*Richtungsquantelung*) is more meaningfully translated as ‘quantization of direction’.

Enter Otto Stern (1888–1969). He had trained primarily as a theorist, and before World War I he had been Einstein’s assistant, first in Prague and then in Zurich. During the turmoil that followed the 1918 armistice and Germany’s ousting of its monarchy, Stern took on an adjunct faculty position at the recently established University of Frankfurt, working under theorist Max Born. In 1919, Stern became aware of the *Richtungsquantelung* question, and during a sleepless night he realized that it could be tested with a hot new technique³.

Breakthroughs in vacuum technology in the previous decade, and in particular the diffusion-pump principle introduced by Wolfgang Gaede in 1915, had enabled researchers to reach the regime where the mean free path of molecules was longer than the dimensions of the device. This made possible the technique of molecular beams, and suddenly turned many *Gedanken* experiments into practical possibilities. As Stern would later say in his [Nobel lecture](#): “It enables us to make



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measurements on isolated neutral atoms or molecules with macroscopic tools.”

When it came to quantization, Stern’s idea was that in the Bohr–Sommerfeld theory, the orbit of the silver atom’s outermost electron should produce a current, and with it a net magnetic moment. An inhomogeneous magnetic field should then exert a force on the atoms that depends on the angle between this magnetic moment and the field. As a consequence, a beam of silver atoms going through such a magnetic field would be deflected by an amount proportional to (the cosine of) that angle. If *Richtungsquantelung* was real, the angle could only take two values, and the deflection would in effect split the beam into two; but if it was not, the beam would just end up spread around a middle point.

A challenging experiment

To help with the work, Stern recruited Walther Gerlach (1889–1979), who had joined the University of Frankfurt in 1920. Funding the experiment was another matter. The long war and an onerous peace treaty had brought Germany to the edge of financial ruin. Born helped out by sharing part of the proceeds of his hugely popular public lectures on general relativity, a theory that had just gained front-page prominence following observations of its effects on the 1919 solar eclipse. New York City banker Henry Goldman of Goldman–Sachs fame is said to have provided additional financial help.

Stern and Gerlach knew that to be able to see a split beam, they would have to stretch the state of the art in many respects, says Bretislav Friedrich, a molecular physicist who has written extensively about the experiment. “They knew what they were up against,” he says. The expected splitting was on the order of 0.2 mm, which required the various components of the apparatus to be aligned to better than 10 μm (REF.⁴).

One of Gerlach’s tasks was to build an electromagnet that could provide a strong vertical gradient. On one side of the magnet, the atom beam was provided by an oven that heated silver metal to more than 1,000 °C and emitted a beam of monoatomic silver vapour through a hole. On the other side of the magnet, the silver deposited onto a glass plate. The whole apparatus had to withstand

strong temperature gradients, as the Gaede pump had to be kept at liquid-nitrogen temperatures. (The recent invention of borosilicate glass, which is highly resistant to temperature gradients, might have helped.)

Progress was laborious. Achieving the required vacuum required hours of pumping. Then, on most days, some part or other of the glass vacuum chamber would break, and the whole process had to begin again. It didn’t help that the beam had low intensity, and took hours to deposit appreciable amounts of silver.

Towards the end of 1921, Stern took up a professorship in Rostock, on the Baltic coast. With no money to start a lab anew, he continued the silver-beam collaboration with Gerlach, whom he left to do the day-to-day work while he stayed abreast from a distance.

Gerlach finally saw the splitting on the night of 7 February 1922. On the morning of the 8th, he sent a telegram to his collaborator, saying, “Bohr was right after all.” That same day, Gerlach wrote a postcard to Bohr with a photograph of the splitting — taken through a microscope — and a message of congratulations (pictured).

A matter of interpretation

In reality, the experiment had been a triumph, but for very different reasons than physicists realized at first. As the quantum mechanics revolution established only a few years later, silver’s unpaired electron is in the spherically symmetric 5s orbital, in which the electron’s wavefunction is all over the place and carries zero orbital angular momentum. As such, it produces no magnetic moment. But the electron itself has an intrinsic angular momentum, and an associated intrinsic magnetic dipole moment.

With an estimated accuracy of 10%, Stern and Gerlach had measured the splitting predicted by Bohr–Sommerfeld theory, according to which the electron had an angular momentum of one Planck unit, $\hbar$. This turned out to be correct, but — again — for the wrong reason. When measured along a given axis, the electron has an angular momentum of $\hbar/2$, not $\hbar$, something that follows from the 1928 relativistic wave equation of Paul Dirac. But as Dirac also showed, the electron has (neglecting higher-order QED terms) a ‘gyromagnetic factor’ $g=2$, which in essence means that its spin generates twice the expected magnetic moment. Together, the 2 and the $\frac{1}{2}$ conspired to give the expected deviation.

In the full quantum mechanical interpretation of the experiment, each electron comes out of the oven in a random spin state. Its spin degree of freedom is encoded not as a vector in 3D space, but as an ‘internal’ state in a Hilbert space with two complex dimensions. (It is often said that spin can only be ‘up’ or ‘down’, but technically that is only true after measurement.) As Pauli showed in 1927, measurement of angular momentum along the vertical direction corresponds to an operator with eigenvalues $\pm\hbar/2$.

The magnetic moment is never fully aligned with the magnetic field: the electron’s ‘total’ angular momentum vector, to the extent that it makes sense to talk about it, is always at an angle to the vertical direction. In practice, only one vector component can be measured, because the three components are quantum mechanical

operators that do not commute: the Heisenberg uncertainty principle implies that when one component of the vector is known — in this case, the vertical one — the other two are completely undetermined.

But what constitutes a measurement of the spin being up ($+\hbar/2$) or down ($-\hbar/2$)? According to quantum mechanics, in the initial state, the electron is in a random superposition of the up and down state. That means the electron is in a superposition of two wavefunctions, corresponding to the two spin states, and initially these have the same probability distribution across space. As the atom goes through the magnetic field, these two states become displaced from each other: in other words, spin becomes entangled with position. It is only when the wavefunctions arrive at the screen and the atom is deposited that the electron ‘decides’ which spin state it was in — in other words, it is at this point the measurement is made.

Modern-day tests

It was not until the 21st century that clear evidence emerged that indeed, the mere act of separating two spin states in space does not constitute a measurement, and it is in fact reversible⁵. Physicist Ron Folman presented the results in September 2019 at a Frankfurt conference that celebrated the Stern–Gerlach experiment⁶. “I was very happy to be there and show, for the first time after 100 years, that their experiment was what textbooks claimed it is,” says Folman.

In the experiment, Folman’s colleague Yair Margalit suspended a Bose–Einstein condensate (BEC) under

an atom chip and then let it fall by gravity. During that fall, Margalit first separated the BEC’s two spin states horizontally, then reversed that motion and made them recombine. An interference pattern showed that coherence had been maintained, similar to what happens in double-slit experiments, says Margalit. “We saw these beautiful matter wave interferometry fringes.”

Many physicists over the decades had doubted that such a ‘Stern–Gerlach interferometer’ could ever work: as Julian Schwinger put it, maintaining coherence after the beam has split and recombined would be like trying to put Humpty Dumpty back together. But Folman says that the experiment proved many sceptics wrong. In the future, the technique could be used to detect gravitational waves and to probe quantum gravity.

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